

AI-Based Smart Digital Lock System Using VHDL

Project Title

AI-Based Smart Digital Lock System Using VHDL with Security Alert

Introduction

The AI-Based Smart Digital Lock System is an advanced FPGA security project developed using VHDL. Unlike normal digital lock systems, this project includes intelligent security features such as wrong-attempt detection, smart alarm control, automatic lock timeout, and security alert indication.

The system uses FSM (Finite State Machine) architecture for password verification and lock management. It improves security by detecting suspicious access attempts and activating protection mechanisms automatically.

This project is unique because it combines FPGA design concepts with smart security logic, making it more advanced than a basic password lock system.

Objectives

- Design a smart digital lock system using VHDL
 - Implement FSM-based password authentication
 - Detect multiple wrong password attempts
 - Activate smart security alert system
 - Interface keypad and 7-segment display
 - Simulate using ModelSim/Vivado
 - Deploy on FPGA board
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Unique Features

- Smart password authentication
- AI-like suspicious activity detection
- Automatic alarm activation
- Auto-lock timeout feature
- Wrong attempt counter
- Emergency master reset
- LED-based security indication
- 7-segment display support

- FPGA implementation ready

Technologies Used

- VHDL
- FPGA
- FSM Design
- ModelSim
- Vivado

Hardware Requirements

- FPGA Board (Spartan/Artix)
- 4x4 Keypad
- LEDs
- 7-Segment Display
- Buzzer
- Push Button

Software Requirements

- Vivado
- ModelSim
- VHDL Compiler

System Architecture

```

graph LR
    subgraph "5u17y4"
        direction TB
        KIM[Keypad Input Module]
        PVP[Password Verification]
        FSC[FSM Security Control]
        UAS[Unlock Alarm System]
        KIM --> PVP
        PVP --> FSC
        FSC --> UAS
    end

```

```

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```

```

# Working Principle

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## Step 1: Password Entry

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User enters a password through keypad interface.

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```

## Step 2: Verification
Entered password is compared with stored password.

## Step 3: Correct Password
If password is correct:

- Door unlocks
- Green LED turns ON
- Display shows OPEN

## Step 4: Wrong Password
If password is incorrect:

- Red LED turns ON
- Attempt counter increases

## Step 5: Smart Security Detection
After multiple wrong attempts:

- Alarm activates automatically
- System blocks further entries temporarily
- Security warning LED blinks

## Step 6: Auto Lock
After a fixed delay:

- Door automatically locks again

## Step 7: Master Reset
Administrator can reset the entire system using reset button.

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# FSM States

| State | Description |
|---|---|
| IDLE | Waiting for user input |
| INPUT | Accepting password |
| VERIFY | Checking password |
| OPEN | Door unlocked |
| ERROR | Invalid password |
| ALERT | Suspicious activity |
| LOCKDOWN | System blocked |
| RESET | System reset |

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```

Advantages

- High security
- Smart protection mechanism
- Real-time FPGA operation
- Beginner-friendly implementation
- Low hardware cost
- Unique internship project

Applications

- Smart homes
- Office security
- Bank locker systems
- Industrial security
- Hostel room security
- IoT security systems

Simulation

The project can be simulated using:

- ModelSim
- Vivado Simulator

Waveforms verify:

- FSM transitions
- Password matching
- Alarm activation
- Auto-lock operation

Future Enhancements

- RFID card support
- Fingerprint authentication
- Bluetooth mobile unlocking
- GSM alert message system
- Face recognition integration
- Cloud-based monitoring

Expected Output

- Correct password unlocks system
- Wrong attempts trigger alert

- Alarm activates after suspicious activity
- System auto-locks after timeout
- LEDs indicate system status

GitHub Repository Structure

```
```text id="u8r0u5"
```

```
AI-Smart-Digital-Lock-System/
```

```
|
```

```
|— VHDL_Code/
```

```
| |— smart_lock.vhd
```

```
| |— keypad_interface.vhd
```

```
| |— display_driver.vhd
```

```
|
```

```
|— Testbench/
```

```
| |— tb_smart_lock.vhd
```

```
|
```

```
|— Simulation/
```

```
| |— waveform_images
```

```
|
```

```
|— Images/
```

```
| |— block_diagram.png
```

```
|
```

```
|— Report/
```

```
| |— project_report.pdf
```

```
|
```

```
|— README.md
```

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## Conclusion

The AI-Based Smart Digital Lock System Using VHDL is a unique FPGA security project that combines FSM logic, password authentication, and intelligent security features. The project demonstrates practical VLSI and FPGA concepts while providing an advanced and professional internship-level implementation.